

Thomas Evans

587-974-3501 | tevans3@ualberta.ca | tevans-3.github.io | github.com/tevans-3 | LinkedIn

EXPERIENCE

University of Alberta, Department of Family Medicine

Edmonton, AB

Software Developer

April 2025 – Present

- Designed, deployed, and maintained a Flask/SQLite application serving as internal system-of-record and HR management platform for the department's clinical faculty complement.
- Developing a clinic analytics platform: SQL Server database, C# ASP.NET Core MVC application, Razor/JS frontend, Python ETL pipelines, and AI-assisted querying, serving multiple clinic sites.
- Designed and built a SQL Server system with temporal tables for tracking historical data changes and point-in-time analysis; built Python and SQL ETL/ELT pipelines to migrate legacy spreadsheet and SharePoint data.
- Designed an English-to-SQL conversion layer: used Gemini's API to generate SQL from executive natural-language input against a fully isolated read-only database replica; secured the conversion layer using ScriptDom to AST-parse AI-generated SQL and reject non-SELECT query patterns.
- Worked closely with non-technical stakeholders to gather requirements: led a requirements gathering workshop for 16 end-users; conducted in-person site observations of clinic staff workflows; delivered technical design documentation.

Research Assistant

May 2025 – Jul 2025

- Managed data collection, analysis, and study design for usability study comparing faculty reporting systems.
- Implemented PDF report generation and statistical analysis (ordinal logistic regression, sign tests, Cronbach's α , PCA).
- Adapted System Usability Scale for domain context; wrote technical report; lead author on poster (ACFP 2026); oral presenter at 2026 FM Research Day.

Program Assistant

Sep 2022 – April 2025

- Prototyped Python tool using Gemini API for data deidentification; automated administrative tasks with Python and JavaScript.

PROJECTS

Open Source Contributions

Rust. Active contributor to [Bevy Engine](#)

- Migrating engine examples to the new Feathers widget set.

Synesthesia

JavaScript, React. [Front page Hacker News, Jan 2026. Try it here](#)

- Color-picker sonification tool. Answers the all-important, widely-asked question: 'What does a color-picker sound like?'
- Designed an algorithm mapping hexcodes to chords in the 24 tone-equal-temperament scale.
- Wrote bit-twiddling transformation logic to implement hexcode-to-chord morphisms.
- Implemented realtime audio synthesis with Attack-Decay-Sustain-Release envelopes using Tone.js.
- Used React hooks e.g. `memo`s and `refs` to reuse Tone.js nodes, saving overhead on per-render regenerations.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript/TypeScript, SQL, Rust

Frameworks: .NET Core, Entity Framework, React, Flask

Infrastructure: SQL Server, Docker, Git, Linux

EDUCATION

University of Alberta

Edmonton, AB

BSc, Specialization in Computing Science

Part-time; Expected Fall 2027

Selected Coursework: Algorithms, Operating Systems, Parallel & Distributed Systems, Cryptography & Digital Privacy